

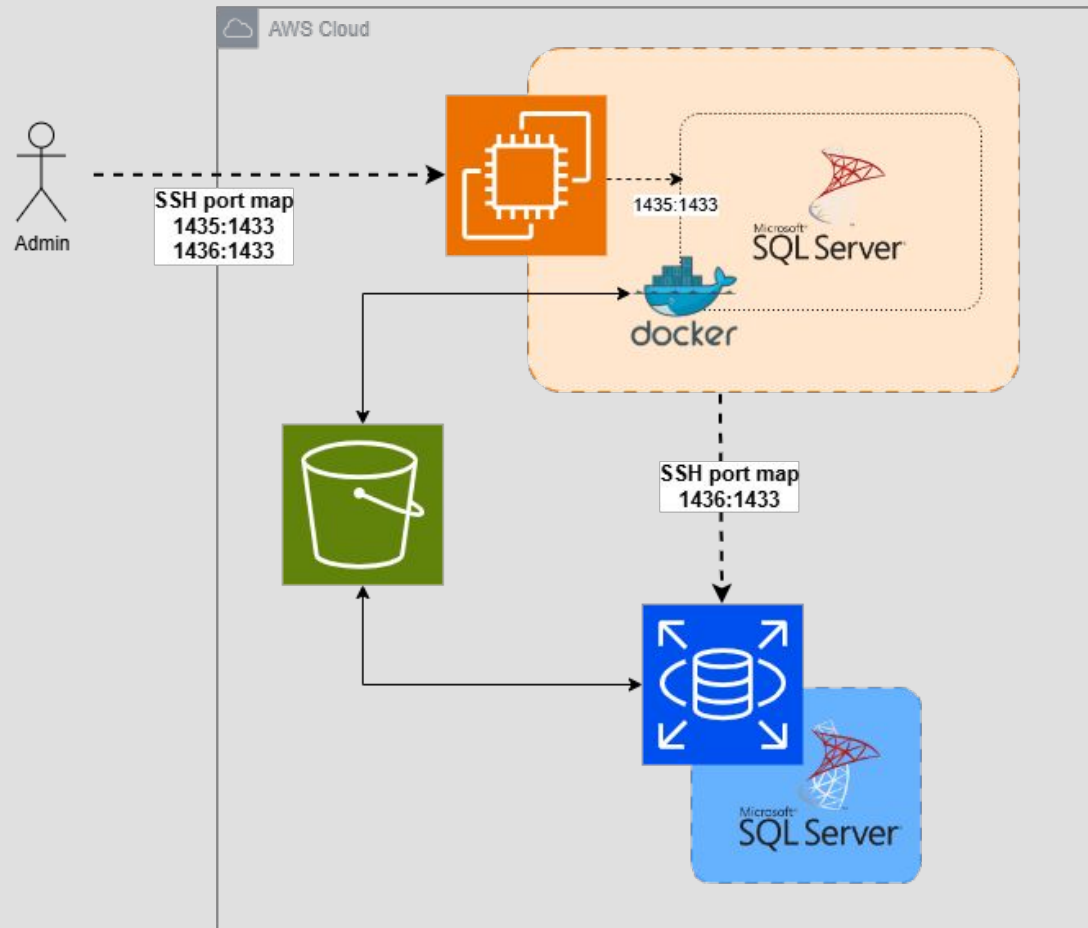


# EC2 + Docker + RDS + Option Group + S3 + MS SQL Server



## OBJETIVOS

- Rodar MS SQL Server em Docker
- Realizar migração de banco local para RDS
- Realizar migração de banco no RDS para local



The screenshot displays the AWS Data Studio interface. On the left, the 'Databases' panel shows a connection to 'RDS SQL Server' on 'localhost:1436'. Under this connection, the 'AdventureWorks' database is listed with a size of 272M. The main editor window shows a SQL script in 'Script-11' with the following commands:

```
exec msdb.dbo.rds_restore_database  
@restore_db_name='AdventureWorks',  
@s3_arn_to_restore_from='arn:aws:s3:::asfags-bucket-mssql/AdventureWorks2022.bak';  
  
exec msdb.dbo.rds_task_status @task_id=4;
```

Below the script, the 'Results' panel shows a single record from a query. The record is highlighted with a red box and contains the following data:

| task_id | AZ task_type | AZ database_name | % complete | duration(mins) | AZ lifecycle | task_info                                |
|---------|--------------|------------------|------------|----------------|--------------|------------------------------------------|
| 4       | RESTORE_DB   | AdventureWorks   | 100        | 2              | SUCCESS      | [2026-04-19 20:08:22.433] Task execution |

The bottom status bar indicates 'Databases - Server databases' and 'BRT en'.

Restaurando database no RDS a partir do backup salvo no bucket do S3

The screenshot displays the SQL Server Enterprise Manager interface. On the left, the 'Databases' pane shows the 'AdventureWorks' database. The main window is split into two panes. The top pane shows a query window with the following SQL code:

```
select * from AdventureWorks.Person.EmailAddress ea
```

The bottom pane shows the 'Results 1' window, displaying a table with the following data:

| BusinessEntityID | EmailAddressID | EmailAddress                 | rowguid                              | ModifiedDate            |
|------------------|----------------|------------------------------|--------------------------------------|-------------------------|
| 1                | 1              | ken0@adventure-works.com     | 8A1901E4-671B-431A-871C-EADB2942E9EE | 2009-01-07 00:00:00.000 |
| 2                | 2              | terri0@adventure-works.com   | B5FF9EFD-72A2-4F87-830B-F338FD4D162  | 2008-01-24 00:00:00.000 |
| 3                | 3              | roberto0@adventure-works.com | C8A51084-1C03-4C58-A8B3-55854AE7C499 | 2007-11-04 00:00:00.000 |
| 4                | 4              | rob0@adventure-works.com     | 17703ED1-0031-4B4A-AFD2-77487A556B3B | 2007-11-28 00:00:00.000 |
| 5                | 5              | gail0@adventure-works.com    | E76D2EA3-08E5-409C-BBE2-5DD1CDF89A3B | 2007-12-30 00:00:00.000 |

The second pane shows a query window with the following SQL code:

```
update AdventureWorks.Person.EmailAddress  
set emailaddress = 'test@test.com'  
where emailaddressid = 1;
```

The bottom pane shows the 'Results 1' window, displaying a table with the following data:

| BusinessEntityID | EmailAddressID | EmailAddress                 | rowguid                              | ModifiedDate            |
|------------------|----------------|------------------------------|--------------------------------------|-------------------------|
| 1                | 1              | test@test.com                | 8A1901E4-671B-431A-871C-EADB2942E9EE | 2009-01-07 00:00:00.000 |
| 2                | 2              | terri0@adventure-works.com   | B5FF9EFD-72A2-4F87-830B-F338FD4D162  | 2008-01-24 00:00:00.000 |
| 3                | 3              | roberto0@adventure-works.com | C8A51084-1C03-4C58-A8B3-55854AE7C499 | 2007-11-04 00:00:00.000 |

Alterando dado da tabela EmailAddress para posterior restore em ambiente local

The screenshot displays the AWS Data Studio interface. On the left, the 'Databases' pane shows a connection to 'RDS SQL Server' at 'localhost:1436', with a tree view including 'AdventureWorks' (272M), 'master' (6.4M), 'rdsadmin' (16M), 'Security', and 'Administer'. The main editor shows two SQL scripts. Script-11 contains the following T-SQL code:

```
exec msdb.dbo.rds_backup_database  
@source_db_name='AdventureWorks',  
@s3_arn_to_backup_to='arn:aws:s3:::asfags-bucket-mssql/AdventureWorks-custom.bak',  
@number_of_files=1;  
  
exec msdb.dbo.rds_task_status @task_id=5;
```

Script-12 is currently empty. Below the scripts, the 'Results 1' pane shows a table with the following data:

| task_id | AZ task_type | AZ database_name | 123 % complete | 123 duration(mins) | AZ lifecycle | AZ task_info | last_updated                         |
|---------|--------------|------------------|----------------|--------------------|--------------|--------------|--------------------------------------|
| 5       | BACKUP_DB    | AdventureWorks   | 100            |                    | 2            | SUCCESS      | 2026-04-19 20:14:2026-04-19 20:15:22 |

The bottom status bar indicates 'BRT en Writable', 'Smart Insert', and a time of '6:42 [41]'.

Exportando database para bucket do S3

Objects (2)



Copy S3 URI

Copy URL

Download

[Open ↗](#)

Delete

Actions ▼

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

| <input type="checkbox"/> | Name ▲                                    | Type ▼ | Last modified ▼                      | Size ▼   |
|--------------------------|-------------------------------------------|--------|--------------------------------------|----------|
| <input type="checkbox"/> | <a href="#">AdventureWorks-custom.bak</a> | bak    | April 19, 2026, 17:14:23 (UTC-03:00) | 198.5 MB |
| <input type="checkbox"/> | <a href="#">AdventureWorks2022.bak</a>    | bak    | April 17, 2026, 20:54:35 (UTC-03:00) | 200.1 MB |

Validando envio do arquivo no bucket do S3



```
admin@ip-172-31-18-179: ~
File Edit View Search Terminal Help

admin@ip-172-31-18-179:~$ docker run --name sql -e "ACCEPT_EULA=Y" -e "MSSQL_SA_PASSWORD=P@ssword3"
-p 1433:1433 -d mcr.microsoft.com/mssql/server:2022-latest
31233454cf8e6ba0403903b7920d6b0bf345bcc16187249ed33e51ce06f356e3
admin@ip-172-31-18-179:~$ aws s3 cp s3://asfags-bucket-mssql/AdventureWorks-custom.bak .
download: s3://asfags-bucket-mssql/AdventureWorks-custom.bak to ./AdventureWorks-custom.bak
admin@ip-172-31-18-179:~$ docker cp ./AdventureWorks-custom.bak sql:/var/opt/mssql/data/
Successfully copied 208MB (transferred 208MB) to sql:/var/opt/mssql/data/
admin@ip-172-31-18-179:~$
```

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Databases' tree is expanded, showing 'AdventureWorks' (272M) and 'master' (6.2M). The 'EC2 Docker SQL Server' connection is selected. In the center, the 'Script-13' window displays the following T-SQL commands:

```
RESTORE FILELISTONLY FROM DISK = 'AdventureWorks-custom.bak'

GO
RESTORE database AdventureWorks
From disk = 'AdventureWorks-custom.bak'
with
move 'AdventureWorks2022' to '/var/opt/mssql/data/AdventureWorks2022'
move 'AdventureWorks2022_log' to '/var/opt/mssql/data/AdventureWorks2022_log'
GO
```

An arrow points from the 'Script-13' window to the 'Output' window on the right. The 'Output' window shows the following message:

```
Processed 25392 pages for database 'AdventureWorks', file 'AdventureWorks2022'
Processed 1 pages for database 'AdventureWorks', file 'AdventureWorks2022_log'
RESTORE DATABASE successfully processed 25393 pages in 3.428 seconds
```

At the bottom, the 'Statistics 1' window shows the following data:

| Name         | Value  |
|--------------|--------|
| Updated Rows | -1     |
| Execute time | 4.235s |

Baixando backup do S3 -> copiando para container MSSQL -> Restore database



The screenshot displays the SQL Server Enterprise Manager interface. On the left, the 'Databases' tree shows the 'AdventureWorks' database selected. The 'Person' schema is expanded, and the 'EmailAddresses' table is highlighted. The 'Query Editor' window shows a SQL query: `select * from AdventureWorks.Person.EmailAddress ea`. The 'Results' pane shows the execution of this query, displaying a table with 8 rows. The first row is highlighted, showing the email address 'test@test.com'.

| BusinessEntityID | EmailAddressID | EmailAddress                 | rowguid                              | ModifiedDate            |
|------------------|----------------|------------------------------|--------------------------------------|-------------------------|
| 1                | 1              | test@test.com                | A1901E4-671B-431A-871C-EADB2942E9EE  | 2009-01-07 00:00:00.000 |
| 2                | 2              | terri0@adventure-works.com   | B5FF9EFD-72A2-4F87-830B-F338FDD4D162 | 2008-01-24 00:00:00.000 |
| 3                | 3              | roberto0@adventure-works.com | C8A51084-1C03-4C58-A8B3-55854AE7C499 | 2007-11-04 00:00:00.000 |
| 4                | 4              | rob0@adventure-works.com     | 17703ED1-0031-4B4A-AFD2-77487A556B3B | 2007-11-28 00:00:00.000 |
| 5                | 5              | gail0@adventure-works.com    | E76D2EA3-08E5-409C-BBE2-5DD1CDF89A3B | 2007-12-30 00:00:00.000 |
| 6                | 6              | josef0@adventure-works.com   | A9C4093A-4F4A-4CAD-BBB4-2C4E920BACCB | 2013-12-16 00:00:00.000 |
| 7                | 7              | dylan0@adventure-works.com   | 70429DE4-C3BF-4F19-A00A-E976C8017FB3 | 2009-02-01 00:00:00.000 |
| 8                | 8              | diane1@adventure-works.com   | 37F02A87-058D-49F8-A20D-965738B0A71F | 2008-12-22 00:00:00.000 |

Validando que o backup restaurado contém a alteração realizada no RDS